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ORIGINAL RESEARCH

Determinants of risk factor target attainment after myocardial infarction: the Perfect-CR study

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Aims How individual components of cardiac rehabilitation (CR) programmes predict attainment of risk factor targets after myocardial infarction (MI), and their predictive strength relative to patient characteristics, remain unclear. We aimed to identify organizational and patient-level predictors of risk factor target attainment at 1-year post-MI.

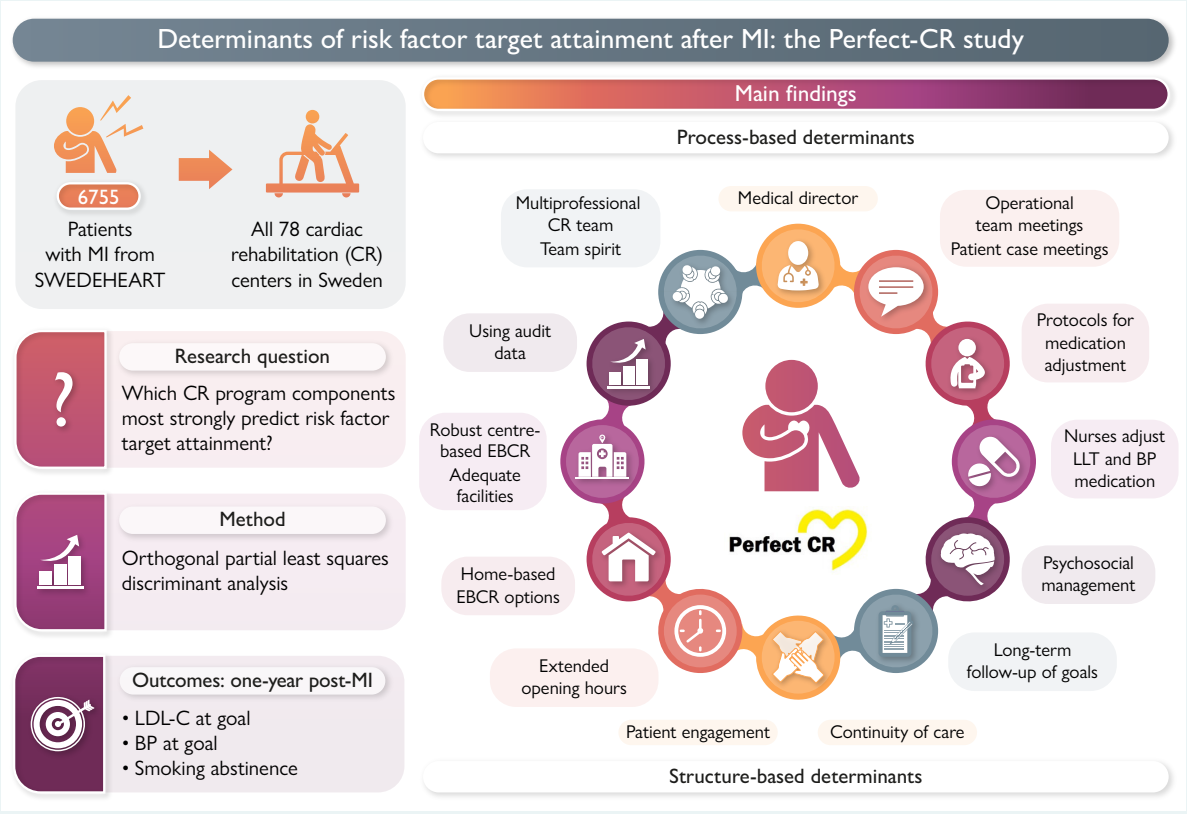
Methods and results Organizational structure and processes followed at all 78 CR centres in Sweden were surveyed and merged with patient-level registry data ($n = 6755$). Predictors of attaining low-density lipoprotein-cholesterol (LDL-C) and blood pressure (BP) targets and smoking abstinence were identified using orthogonal partial least squares discriminant analysis, generating variables of importance for the projection (VIP) values. Variables with VIP > 0.8 were considered meaningful. The strongest predictors for attaining LDL-C and BP targets included availability of psychosocial management, medication adjustment protocols for nurses, continuity in nurse-patient contact, extended opening hours and adequate facilities, and having a medical director at the CR centre. The strongest patient-level predictors of attaining LDL-C and/or BP targets included low baseline LDL-C and having no history of hypertension, respectively, and the patient having participated in exercise-based CR. For smoking abstinence, the strongest organizational predictors were availability of psychosocial management, having a medical director, prescribing nicotine replacement therapy and varenicline, and self-reported team spirit. Patient-level predictors included having participated in exercise-based CR and group education.

Conclusion We identified multiple CR organizational predictors of attaining key risk factor targets after MI. These results may influence the future design of comprehensive CR programmes.

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Graphical abstract



BP, blood pressure, CR, cardiac rehabilitation, EBCR, exercise-based cardiac rehabilitation; LDL-C, low-density lipoprotein cholesterol; LLT, lipid-lowering treatment; MI, myocardial infarction.

Keywords Cardiac rehabilitation • Myocardial infarction • Risk factors • Secondary prevention

Novelty

- The study identifies multiple CR programme components that predicted target attainment for LDL-C, blood pressure, and smoking abstinence post-MI, ranking these by predictive strength, also relative to patient-level predictors.
- The results can contribute to optimally organizing CR programmes, making the best possible use of limited human and economic resources.

Introduction

Cardiac rehabilitation (CR) is a multidisciplinary intervention with clearly defined core components, including patient assessment, management of cardiovascular risk factors, physical activity and dietary counselling, prescription of exercise training, psychosocial management, and vocational support.[1,2] CR carries the highest possible recommendation in the current European guidelines on cardiovascular disease (CVD) prevention[3] and evidence from meta-analyses has affirmed its effectiveness.[4–6]

However, considerable heterogeneity in CR service delivery exists across programmes, including variation in core components such as risk factor management, exercise training, psychosocial support, and follow-up.[4,6] While the framework for optimal CR is well described, the relative contribution of individual components to achieving risk factor targets remains unclear.[1] Identifying predictors of target attainment may help explain this variation and inform optimization of CR programme delivery.

In the repeated cross-sectional EUROASPIRE (European Action on Secondary and Primary Prevention through Intervention to Reduce Events) surveys,[7] target attainment for key risk factors varied considerably between programmes.[8–10] Similar outcome variation has repeatedly been observed in the nationwide SWEDEHEART (Swedish Web-System for Enhancement and Development of Evidence-Based Care in Heart disease Evaluated According to Recommended Therapies) quality registry, that records individual-level baseline characteristics, treatments, and follow-up data of all patients admitted with a myocardial

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infarction (MI) to coronary care units in Sweden.[11] In the latest annual report from 2024, the proportion of patients that attained the set low-density lipoprotein-cholesterol (LDL-C) target varied between 36% and 89% across centres, for systolic blood pressure (BP) attaining target varied between 41% and 100%, and for smoking abstinence the proportion varied between 14% and 100%.[11] Whether heterogeneity in CR delivery can explain some of this variation is unknown.

Through combining unique survey data from all CR centres in Sweden with patient-level data from SWEDEHEART, the present study aimed to identify organizational and patient-level predictors for attaining treatment targets for LDL-C, BP, and smoking abstinence at 1-year post-MI in a nationwide cohort of patients who participated in CR.

Methods

This was an observational survey- and registry-based study. The study complies with the Declaration of Helsinki was reviewed and approved by the Ethics Committee at Lund University (2018/55). The need for signed consent by patients for inclusion in Swedish quality registries has collectively been waived. Upon hospital admission patients with MI are informed verbally and in writing by a nurse or physician about SWEDEHEART.

Organizational centre-level survey data

Data on CR organization were derived from the survey-based Perfect-CR study, evaluating service delivery at Swedish CR centres. Details of the study procedure have previously been published.[12] In short, a web-based questionnaire was sent out in November 2016 to all 79 CR centres in Sweden reporting to SWEDEHEART at the time (95% of all CR centres in Sweden).[13] The respondents were instructed to provide answers that reflected the everyday operation at the centre during 2016. The response rate was 100% and missing data was minimal. One centre shut down weeks after the survey was conducted and was excluded from further analysis.[12] In total, 71 organizational variables representing guidelines-recommended standards of CR delivery were used in the current analysis. The variables included *Structure-based metrics*, *Process-based metrics*, and *Quality metrics*, categorized as suggested by the European Association of Preventive Cardiology.[1] All variables and their definitions are listed in Supplementary material online, *Table S1*.

Patient-level registry data

The patient population in the current study was defined by the discharge date after hospital admission (01.11.2015 to 31.10.2016). The date range was selected by matching the time when patients attended CR (the first 3–6 months after discharge) with the time-period covered by Perfect-CR survey answers. Inclusion criteria were: (i) discharged alive after suffering type-1 MI, (ii) age 18–74 years, and (iii) attending a routine 1-year follow-up CR visit. Data were extracted from the SWEDEHEART and Statistics Sweden registries. Twenty-six patient-level variables were used in the current analysis (see Supplementary material online, *Table S2*). Geographical distance (kilometres) to the CR centre was estimated using co-ordinates of the central point of each patient´s postal code as a proxy for their address, from which the driving distance to the CR centre was calculated by a commercially available algorithm using Google Maps. Patient-level data were linked using the unique Swedish personal identification number allocated to every Swedish citizen. Organizational data were then merged with patient-level data based on which CR centre each patient had their follow-up.

Study outcomes

The study outcomes were attainment of target levels at 1-year after MI for three primary risk factors managed in CR: LDL-C, BP, and smoking abstinence (among smokers at baseline). Outcome variable data were retrieved from SWEDEHEART. LDL-C and BP target attainment was defined in concurrence with guidelines followed at the time of the study: LDL-C < 1.8 mmol/L (yes/no) and BP <140/90 mmHg (yes/no), as these were considered fulfilled targets at the time of the study.[14] Smoking status was self-reported. Abstinence was defined as not smoking for ≥1 month (yes/no) prior to the follow-up visit.

Statistical methods

Baseline characteristics are presented as means (SD) or medians (IQR) for continuous variables and counts (%) for categorical variables. With the comprehensive high-quality data used in the present study, with overall low missingness, missing data were assumed to be missing at random.[15] *k* nearest neighbour imputation was performed with *k* set low (*k* = 3).[16] Gower distance was applied, allowing for calculating a single distance metric from both categorical and numerical variables. Outcome data were not imputed. A multivariable discriminant analysis was performed by Orthogonal Projection to Latent Structures Discriminant Analysis (OPLS-DA)[17] using the non-linear iterative partial least squares (NIPALS) algorithm that allows for the analysis of wide data matrices. OPLS-DA uses Q^2 (goodness of prediction), calculated using leave-one-out cross-validation (LOOCV) resampling, to determine the number of independent (orthogonal) components to extract. OPLS-DA computes the influence of every X-variable on patient outcome (Y), resulting in variables of importance for the projection (VIP) and their respective loading on the first principal component. VIP is a weighted sum of each individual X-variable in the model, and a measure of how much a specific variable contributes to the overall model. In our study, variables with a VIP value >0.8 and a confidence interval (CI) excluding zero were considered to have influence on the projection and, thus, be predictive of outcomes. To aid in the interpretation of the VIP value, the loading value was used, which indicates whether the variables were negative or positive predictors of the outcome. Thereafter, a set of crude logistic regression models was fitted to estimate the unadjusted association of each predictor with the three outcome variables. Point estimate odds ratios (OR) and 95% CI are presented. Baseline characteristics were calculated using IBM SPSS Statistics for Windows, version 25.0 (IBM Corp, Armonk, NY, USA). The OPLS-DA was performed using SIMCA *P*+, version 15.0.2.0 (Sartorius Stedim Data Analytics AB, MKS, Umeå, Sweden). Covariate imputation and logit modelling were performed in R, version 4.0.3 (The R Foundation for Statistical Computing, Vienna, Austria).

Results

Details of the CR organization at the 78 CR centres included in Perfect-CR have previously been published.[12,18,19] In short, all CR programmes were Phase II, i.e. a supervised outpatient programme. The median (Q1-Q3) programme duration was 6.5 (2.5–12.0) months. Only eight centres (10.3%) offered home-based CR as an alternative to a centre-based programme. Almost all centres (*n* = 76, 97.4%) offered supervised exercise-based CR (EBCR), most commonly including two one-hour sessions per week for a minimum of 12 weeks. Seventy-five out of the 78 centres (96.2%) had a multidisciplinary team consisting of a nurse, a physician, and a physiotherapist. Additionally, a psychologist or social worker and a dietician were a part of the CR team at 71 (91.0%) and 66 (84.6%) of the centres, respectively. The programme was led by a medical director at 59 (75.6%) of the centres. Further details on the CR organization are summarized in Supplementary material online, *Table S3*.

In *Figure 1*, a flowchart of patients in the study is displayed. Patient-level baseline characteristics are displayed in *Table 1*.

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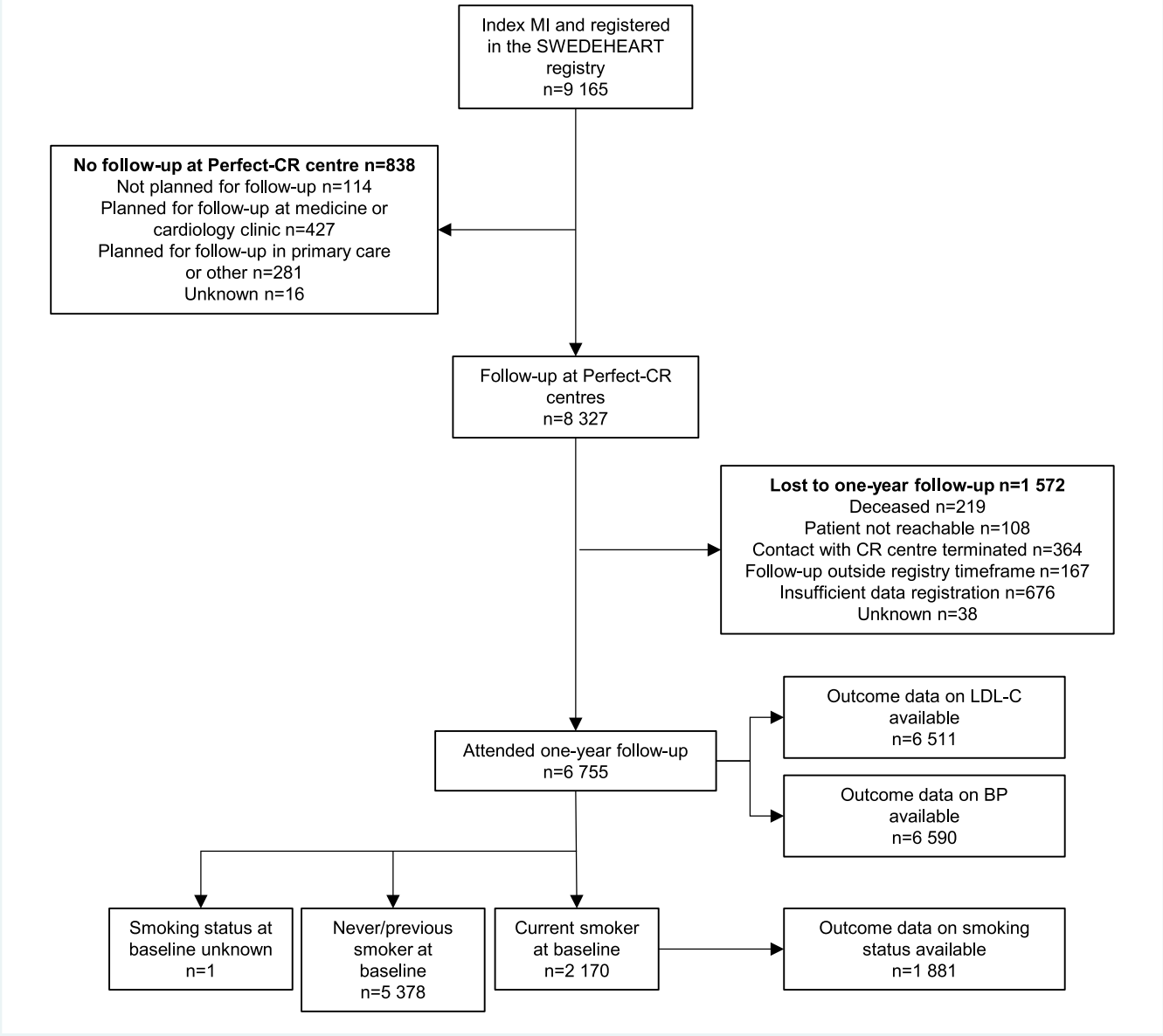


Figure 1 Flow diagram of the study population.

Organizational and patient-level predictors of risk factor outcomes

Variables with a VIP value >0.8 and a CI excluding zero from the OPLS-DA are displayed in *Figure 2* (LDL-C and BP) and *Figure 3* (smoking abstinence). Exact VIP [95% CI] and loading [95% CI] values for all variables, as well as the logistic regression estimates (OR and [95% CI]) are listed in Supplementary material online, *Tables S4–S6*. Main results for each outcome are summarized below.

LDL-C and BP

Out of all 6755 included patients, 6511 (96.4%) had outcome data on LDL-C, of whom 57.8% attained the LDL-C target of <1.8 mmol/L. For BP, 6590 (97.6%) patients had outcome data and 75.8% of these attained the BP target of <140/90 mmHg.

The five strongest *Structure-based metrics* predicting patients being at LDL-C target were: the CR nurses having written protocols on how to adjust secondary prevention medication, the CR team having a medical director responsible for the programme, having written protocols for physician´s coronary care unit discharge consults, the CR nurses independently adjusting the dosage of lipid-lowering medication, and the CR team reporting satisfaction with facilities (*Figure 2*). The five strongest *Structure-based metric* predictors for attaining the BP target were a psychologist or social worker being a part of the CR team, the team reporting satisfaction with facilities, nurses having protocols on how to adjust secondary prevention medication, having a medical director, and having discharge consult protocols at the coronary care unit (*Figure 2*). The strongest *Process-based metrics* for target attainment of LDL-C and BP included: assessing and counselling patients regarding psychosocial health, having flexible opening hours at the CR centre,

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Table 1 Baseline characteristics and CR programme participation for all patients with index MI and those with 1-year follow-up data available

	All patients with index AMI	Patients with 1-year follow-up	*n* (%) missing	All active smokers with index AMI	Active smokers at index with 1-year follow-up	*n* (%) missing
Number of patients	9165	6755		2653	1881	
Demographics						
Age (years)	62.9 ± 8.6	62.6 ± 8.6	0 (0.0)	60.2 ± 8.8	60.0 ± 8.7	0 (0.0)
Sex, male	6874 (75.0)	5145 (76.2)	0 (0.0)	1870 (70.5)	1317 (70.0)	0 (0.0)
Distance between home and CR centre (kilometres)	N/A	21.1 ± 27.0	8 (0.1)	N/A	21.1 ± 28.8	6 (0.3)
Country or region of birth			0 (0.0)			0 (0.0)
Born in Sweden	7340 (80.1)	5508 (81.5)		1884 (71.0)	1361 (72.4)	
Born in Nordic countries (outside of Sweden)	458 (5.0)	314 (4.6)		176 (6.6)	116 (6.2)	
Born in Europe (outside of Nordic countries)	648 (7.1)	453 (6.7)		284 (10.7)	194 (10.3)	
Born outside of Europe	719 (7.8)	480 (7.1)		308 (11.6)	210 (11.2)	
Employment status			837 (12.4)			372 (19.8)
Unskilled workers	2242 (24.5)	1639 (24.3)		753 (35.9)	558 (29.7)	
Skilled workers	1574 (17.2)	1185 (17.5)		451 (21.5)	312 (16.6)	
Assistant non-manual employees	946 (10.3)	725 (10.7)		231 (11.0)	177 (9.4)	
High/intermediate salary employees	2000 (21.8)	1540 (22.8)		319 (15.2)	240 (12.8)	
Self-employed	619 (6.8)	465 (6.9)		145 (6.9)	107 (5.7)	
Other employment	558 (6.1)	364 (5.4)		199 (9.5)	115 (6.1)	
Marital status			1 (0.0)			0 (0.0)
Living with partner	5208 (56.8)	4046 (59.9)		1176 (44.4)	897 (47.7)	
Living alone	3949 (43.1)	2708 (40.1)		1472 (55.6)	984 (52.3)	
Education attainment			57 (0.8)			18 (1.0)
Under 10 years (compulsory school only)	2542 (27.7)	1817 (26.9)		873 (33.4)	613 (32.6)	
10–12 years (upper school)	4449 (48.5)	3287 (48.7)		1335 (51.1)	945 (50.2)	
Over 12 years (college/ university level)	2067 (22.6)	1594 (23.6)		405 (15.5)	305 (16.2)	
Household adjusted income			1 (0.0)			0 (0.0)
Low	1831 (20.0)	1116 (16.5)		785 (29.6)	479 (25.5)	
Medium-low	1833 (20.0)	1310 (19.4)		559 (21.1)	393 (20.9)	
Medium	1830 (20.0)	1379 (20.4)		538 (20.3)	399 (21.2)	
Medium-high	1829 (20.0)	1482 (21.9)		427 (16.1)	341 (18.1)	
High	1834 (20.0)	1467 (21.7)		340 (12.8)	269 (14.3)	
Risk factors, previous diseases, and type of MI						
SBP (mmHg)	150 ± 28	151 ± 27	21 (0.3)	146 ± 29	147 ± 29	7 (0.4)
DBP (mmHg)	88 ± 16	88 ± 16	234 (3.5)	88 ± 17	88 ± 17	79 (4.2)
LDL-C (mmol/L)	3.0 ± 1.1	3.1 ± 1.1	984 (14.6)	3.1 ± 1.1	3.2 ± 1.1	273 (14.5)
BMI (kg/m^2)	28.0 ± 4.7	28.0 ± 4.6	182 (2.7)	27.5 ± 4.9	27.6 ± 4.8	52 (2.8)
History of hypertension	4429 (48.3)	3162 (46.8)	33 (0.5)	1109 (41.8)	782 (41.6)	9 (0.5)
History of CVD	2522 (27.5)	1605 (23.8)	10 (0.1)	613 (23.1)	367 (19.5)	4 (0.2)

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Table 1 *Continued*

	All patients with index AMI	Patients with 1-year follow-up	*n* (%) missing	All active smokers with index AMI	Active smokers at index with 1-year follow-up	*n* (%) missing
History of DM	2126 (23.2)	1304 (19.3)	16 (0.2)	560 (21.1)	312 (16.6)	14 (0.7)
STEMI	3442 (37.6)	2652 (39.3)	0 (0.0)	1240 (46.7)	919 (48.9)	0 (0.0)
LVEF during admission			724 (10.7)			197 (10.5)
Normal (≥50%)	5288 (57.7)	4036 (59.7)		1489 (63.0)	1097 (58.3)	
Mildly reduced (40–49%)	1695 (18.5)	1267 (18.8)		480 (20.3)	346 (18.4)	
Moderately—severely reduced (≤39%)	1093 (11.9)	728 (10.8)		333 (14.1)	241 (12.8)	
Discharge medication						
Platelet inhibitors[a]	9055 (98.8)	6701 (99.2)	0 (0.0)	2629 (99.1)	1870 (99.4)	0 (0.0)
Lipid-lowering therapy[b]	8876 (96.9)	6632 (98.2)	2 (0.0)	2600 (98.0)	1850 (98.4)	0 (0.0)
ACEi or ARB	7766 (84.7)	5809 (86.0)	0 (0.0)	2234 (84.2)	1604 (85.3)	0 (0.0)
Beta blockers	8214 (89.6)	6075 (89.9)	1 (0.0)	2406 (90.7)	1711 (91.0)	0 (0.0)
CR program participation						
Participated in EBCR	3846 (42.0)	3607 (53.4)	26 (0.4)	942 (35.5)	875 (46.6)	7 (0.4)
Participated in group education	3561 (38.9)	3269 (48.4)	35 (0.5)	801 (30.2)	704 (37.6)	8 (0.4)
Participated in professional tobacco counselling	N/A	N/A		382 (14.4)	349 (18.6)	4 (0.2)

Numbers are presented as percentages (%) or means and standard deviations (±SD).
SBP, systolic blood pressure; DBP, diastolic blood pressure; LDL-C, low-density lipoprotein-cholesterol; CVD, cardiovascular disease; HbA1c, haemoglobin A1c; DM, diabetes mellitus, BMI, body mass index; MI, myocardial infarction; STEMI, ST-elevation myocardial infarction; LVEF, left ventricular ejection fraction; ACEi, angiotensin-converting enzyme inhibitors; ARB, angiotensin-II receptor blocker.
[a]Platelet inhibitors included acetylsalicylic acid, clopidogrel, ticlopidine, prasugrel, ticagrelor or other.
[b]Lipid-lowering therapy included statins, ezetimibe, fibrates, and other lipid-lowering therapy.

offering an individual assessment with a physiotherapist before starting EBCR, and during this visit performing a symptom-limited exercise test, the patient receiving written information on lifestyle and risk factors goals at discharge, individually evaluating the patient´s need for a physician consultation during CR follow-up, and continuity in nurse-patient contact and long long-term follow-up of risk factor and lifestyle goals (*Figure 2*). For *Quality metrics*, both patient uptake (proportion of eligible patients attending CR) and the CR centre being situated at a university hospital were identified as predictors of attaining both LDL-C and BP targets. The strongest patient-level predictors for attaining LDL-C targets at one-year were LDL-C at baseline (negative predictor), male sex, having participated in EBCR, and having a history of diabetes mellitus (DM). For attaining the BP target, the strongest patient-level predictors were a history of hypertension and DM (both negative predictors), having participated in EBCR, and having had an ST-elevation MI.

Smoking abstinence

Out of the 6755 patients included, 2170 (32.1%) patients were current smokers at baseline. Outcome data were available for 1881 patients (87.7%), of which 57.2% were abstinent at 1-year post-MI. Out of the *Structure-based metrics*, having a psychologist or social worker in the CR team, having a medical director, reporting good team spirit in the CR team, and using audit data for quality improvement, positively predicted smoking abstinence at one-year post-MI (*Figure 3*). Out of the *Process-based metrics*, CR assessing and counselling patients regarding psychosocial health, duration of the EBCR programme, offering interactive patient education, and physicians prescribing varenicline and recommending the use of nicotine replacement therapy to currently smoking patients were the strongest predictors (*Figure 3*). The strongest patient-level predictors for smoking abstinence included having participated in EBCR and group education, having a history of CVD, DM and hypertension (negative predictors), household disposable income, and living with a partner (*Figure 3*).

Out of the variables identified as predictive by the OPLS-DA, the majority were also statistically significant in the logistic regression for LDL-C (84.0%) and BP (79.2%) (see Supplementary material online, *Tables S4–S5*). There was a larger discrepancy between the OPLS-DA and logistic regression for smoking abstinence (46.8% consistency), with confidence intervals being wide for many variables (see Supplementary material online, *Table S6*). For all outcomes, the exposure showed the same direction of association—positive or negative—in the OPLS-DA and logistic regression models.

Discussion

In this study, using unique nationwide CR organizational and patient-level registry data, we found multiple organizational and patient-level variables to be predictive of patients attaining recommended targets for LDL-C and BP, and of current smokers at baseline being abstinent at 1-year post-MI.

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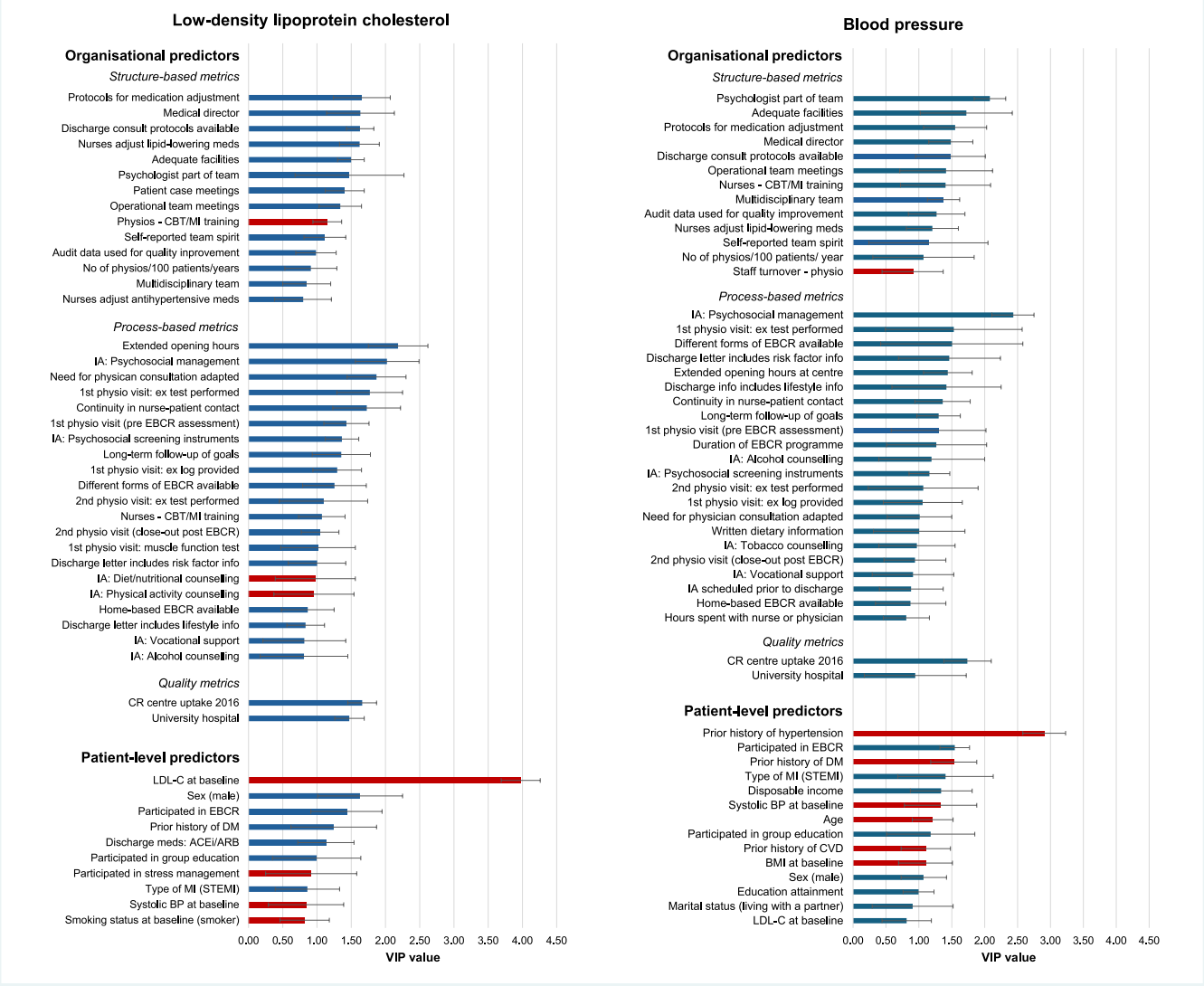


Figure 2 Organizational and patient-level predictors of attaining treatment targets for low-density lipoprotein-cholesterol (<1.8 mmol/L, left panel) and blood pressure (<140/90 mmHg, right panel) at 1-year after myocardial infarction. Negative predictors are shown in red. Only variables with VIP values >0.8 and confidence intervals not including zero are displayed. CR, cardiac rehabilitation; CBT, cognitive behavioural therapy; MI, motivational interviewing, IA, initial assessment; physio, physiotherapist; ex, exercise, EBCR, exercise-based cardiac rehabilitation; LLT, lipid-lowering treatment (statins, ezetimibe, fibrates or other LLT), DM, diabetes mellitus; ACEi, angiotensin converting enzyme inhibitors; ARB, angiotensin II receptor blockers; MI, myocardial infarction; STEMI, ST-elevation MI; LDL-C, low-density lipoprotein cholesterol; BP, blood pressure; BMI, body mass index; CVD, cardiovascular disease.

LDL-C and BP

One of the strongest *structure-based metrics* predicting patients being at target for both LDL-C and BP was to have a medical director. Even though a medical director in the CR team is recommended in international guidelines, supporting evidence is limited.[1,20] Studies from other medical fields have shown that clear leadership is associated with improved patient outcomes and quality of care.[21] Included in the role of a medical director in CR is to ascertain guideline adherence, ensure that care is evidence-based, and to promote quality improvement in everyday clinical practice.[20] Also, including a psychologist in the CR team was a strong predictor for both outcomes. The prevalence of mental health disorders among patients with CVD is high, and interdisciplinary cooperation in patient care is recommended for post-MI patients with mental health disorders.[3]

Several predictors indicating the importance of training and autonomy of CR nurses were identified as meaningful, including nurses having written protocols for medication adjustment, independently adjusting lipid and blood pressure-lowering medication, and having formal training in motivational interviewing. Studies of patients undergoing surgery or having DM have shown that having autonomous nurses was associated with better risk factor outcomes and decreased mortality rates.[22,23] Other structure-based variables of importance were having adequate facilities, operational team meetings, and patient case meetings. Also, using audit data for quality improvement was predictive of both outcomes, strengthening prior conclusions

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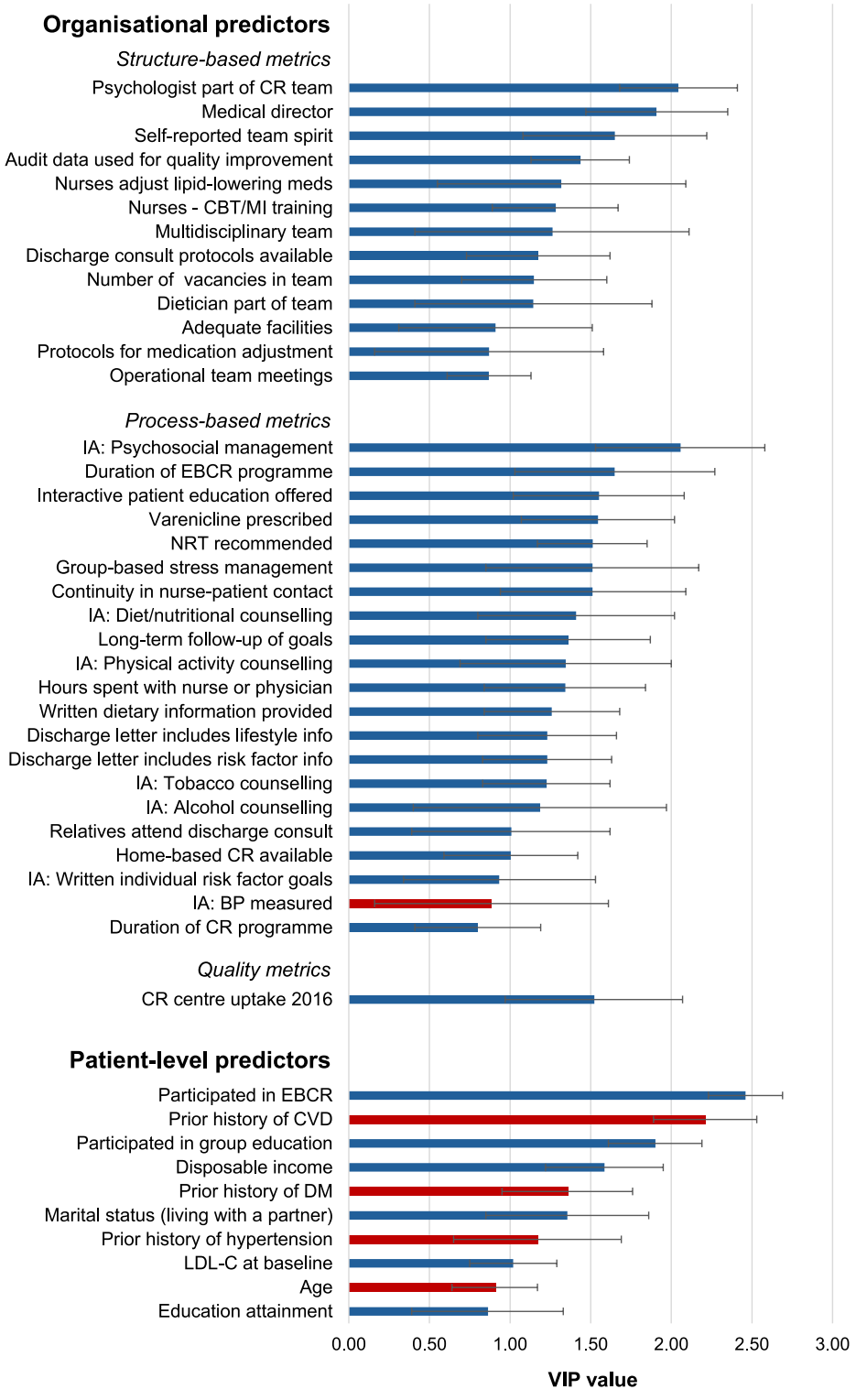


Figure 3 Organizational and patient-level predictors of being abstinent from smoking at 1-year after myocardial infarction. Negative predictors are shown in red. Only variables with VIP values >0.8 and confidence intervals not including zero are displayed. CR, cardiac rehabilitation; IA, initial assessment; EBCR, exercise-based CR; MI, myocardial infarction; STEMI, ST-elevation MI; LDL-C, low-density lipoprotein cholesterol; CVD, cardiovascular disease; DM, diabetes mellitus.

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on the importance of auditing for quality improvement in patient care.[24] Interestingly, the number of nurses on staff was not predictive of attaining LDL-C or BP targets, and hours spent with a nurse and/or physician during follow-up were marginally predictive of attaining BP. Jointly, these results indicate that the composition and competence of the CR team, teamwork, and leadership are more predictive of reaching treatment targets than the number of staff or hours spent with the individual patient. In addition, patient uptake and the centre being situated at a university hospital were predictive of LDL-C and BP. These two predictors may be related since university hospitals are usually situated in densely populated areas, meaning patients have a shorter travel time to the CR centre, which in turn positively affects uptake.[25] University hospitals providing specialized expertise, actively engaging in research and education, might also correlate with better quality of patient care.

Concerning *process-based metrics*, one of the strongest predictors was to have psychosocial management as a part of the initial nurse assessment. Psychosocial risk factors such as low socioeconomic status, social isolation, depression, and anxiety have been shown to worsen the prognosis in patients with established coronary heart disease and act as barriers to lifestyle changes and treatment adherence, decreasing the effects of CR.[26,27] European guidelines recommend that screening for psychosocial risk factors be done by any member of the multidisciplinary team deemed best suited by each institution based on resources and preferences.[1,26] If psychosocial risk factors are identified, the patient should be referred to a psychologist or behavioural expert and as previously stated, having a psychologist as a part of the multidisciplinary team was predictive of both outcomes. Other predictive process-based metrics, including flexible opening hours at the CR centre, continuity in nurse-patient care, long-term follow-up of goals, and the need for physician consultation being adapted to patient needs, reflect a CR organization that provides care tailored to patient needs. Evidence supports the application of individualized CR where the approach to behavioural change is person-centred.[4,6]

Most of the EBCR variables included in the analysis were positive predictors for reaching targets for LDL-C and BP. EBCR participation improves risk factor intensity and reduces hospital readmissions, cardiovascular morbidity, and mortality.[28–30] Our results further support the central role of well-equipped, well-staffed, and flexible EBCR programmes as a part of comprehensive CR, offering individualized consultations with physiotherapists, pre-exercise screening including tests of aerobic and muscular capacity, and a variety of group-training sessions, and the possibility for home-based exercise alternatives.[2]

Dietary and physical activity counselling at the initial assessment, as well as physiotherapists having formal training in motivational interviewing, were negative predictors of LDL-C target attainment at 1-year. The reason for this is ambiguous but could be methodological. As such, there was near-zero variance in physical activity counselling at the initial assessment. Also, the VIP and loading values for all three variables were in the lower range and should be interpreted with caution.

Baseline LDL-C was, not surprisingly, the strongest negative *patient-level predictor* for attaining the LDL-C treatment target. Likewise, a history of hypertension was the strongest negative predictor of reaching BP targets. EBCR participation was a strong patient-level predictor for both outcomes. Also, patients participating in interactive group education were more likely to attain LDL-C and BP targets. Participating in EBCR and patient education events might reflect more engaged patients who are adherent and motivated. However, previous interventional studies have shown that post-MI patients who receive education have fewer cardiovascular events.[31,32]

Smoking abstinence

For *structure-based metrics*, as for LDL-C and BP, having a psychologist in the CR team, having a medical director, self-reported team spirit and using audit data for quality improvement, as well as variables indicating nurse autonomy, were predictors of smoking abstinence at one-year. These predictors indicate the importance of a well-functioning and qualified CR team. Of the *process-based metrics*, varenicline being prescribed by the centre physicians and recommending nicotine replacement therapy were, not surprisingly, strong predictors of smoking cessation. Varenicline has been shown to increase the likelihood of achieving smoking abstinence, but the treatment is highly underutilized in CR.[9,12] Hours spent with a nurse or physician during the first year post-MI were predictive of smoking abstinence, indicating that more face-to-face time during follow-up might be of benefit to smokers. This is corroborated by our data that participation in group education and in EBCR were positive predictors of smoking abstinence, both involving face-to-face contact with the patient. Other *patient-level predictors* for smoking abstinence included variables indicating a favourable socioeconomic status. Negative predictors included age, history of CVD, DM, and hypertension. These variables have previously been identified as predictors for CR attendance.[33] All these attributes are simultaneously non-modifiable by the CR team. The results indicate that smokers are a vulnerable patient group that may benefit even more from person-centred CR than non-smokers.

Strengths and limitations

Our study included data from a nationwide cohort with 100% response rate on the CR centre level and high patient coverage. However, 1572 (18.9%) patients had insufficient data registration or were lost to one-year follow-up, which excludes non-attenders, patients with lower treatment adherence, and patients who died, causing a selection bias and healthy cohort effect. Still, with attendance in CR in Sweden being considerably higher than reported across Europe, generalizability of our results should be higher than in prior observational CR-based studies.[34] Centre level variables reflected the availability of programme components, and participation in some programme components was reported; however, patient level exposure was not guaranteed for all reported organizational centre level variables. Generalizability may be limited due to the age cut-off (18–74 years) as well as due to the use of risk factor targets followed at the time that subsequently have been changed. Smoking abstinence in SWEDEHEART is defined as 1 month (self-report), which might overestimate actual long-term cessation rates. In a global context, Swedish CR programmes are highly functional, limiting the applicability of the results to other less developed healthcare systems. OPLS-DA has the advantage of being able to analyse all variables together, allowing us to look for patterns in prediction and investigate relations between all variables in a single context. Thus, by using OPLS-DA, more variables could be included in the analysis than what is possible in traditional regression models. At the same time, results should be interpreted as exploratory, and patterns of predictors and variable clusters should be interpreted from a bird´s eye view rather than focusing on each single predictor identified. The OPLS-DA approach was complemented with traditional regression analysis estimates per each single predictor, with generally high consistency. We then proceeded to apply human subject matter expertise in interpreting the results. Although NIPALS and, consequently OPLS-DA can overfit to data, we did apply both (a) resampling in the form of LOOCV to counter the risk of overfitting and (b) crude log-linear regression as a sensitivity

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analysis. Neither the OPLS-DA nor the logistic regression accounted for clustering. Aside from modelling parsimony reasons, this was done since the main modelling also included multiple individual-level predictors—in part capturing cluster-level effects by proxy. Finally, being an observational study, no conclusions can be drawn on causality.

Conclusions

We identified multiple organizational predictors for attaining targets for LDL-C BP as well as smoking cessation at 1-year after MI in Swedish patients who had attended a CR programme. Our results lend support to a multiprofessional CR team with strong leadership, interdisciplinary teamwork, regular auditing, flexible and person-centred care being positively associated with beneficial risk factor outcomes. The findings can contribute to optimally organizing CR programme delivery in the era of modern cardiology, making the best possible use of limited human and economic resources.

Supplementary material

Supplementary material is available at *European Journal of Cardiovascular Nursing* online.

Author contributions

Halldora Ögmundsdottir Michelsen (Conceptualization, Investigation [supporting], Formal analysis, Visualization, Writing—review & editing [equal], Writing—original draft [lead]), Peter Henriksson (Data curation, Formal analysis, Methodology [lead], Visualization, Writing—review & editing [equal]), John Wallert (Data curation, Formal analysis, Methodology [supporting], Validation [lead], Visualization, Writing—review & editing [equal]), Maria Bäck (Conceptualization, Writing—review & editing [equal]), Ingela Sjölin (Investigation, Writing—review & editing [equal]), Mona Schlyter (Investigation, Writing—review & editing [equal]), Emil Hagström (Conceptualization, Writing—review & editing [equal]), Anna Kiessling (Conceptualization, Methodology [supporting], Writing—review & editing [equal]), Claes Held (Conceptualization [supporting], Writing—review & editing [equal]), Emma Hag (Conceptualization [supporting], Writing—review & editing [equal]), Lennart Nilsson (Conceptualization [supporting], Writing—review & editing [equal]), Alexandru Schiopu (Supervision [supporting], Writing—review & editing [equal]), and Margret Leosdottir (Conceptualization, Funding acquisition, Investigation, Project administration, Supervision [lead], Formal analysis [supporting], Methodology, Visualization, Writing—review & editing [equal])

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Conflict of interest. The Authors declare that there is no conflict of interest.

Data availability

SWEDEHEART does not allow individual data sharing to a third party. Access to aggregated data might be granted following review by the SWEDEHEART steering committee.

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